

Question No:3

DS	Session	Tree 1	Question Information	Level 1 Challenge 63
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Question description

You are given an $n \times n$ grid representing the map of a forest. Each square is either empty or contains a tree. The upper-left square has coordinates $(1, 1)$, and the lower-right square has coordinates (n, n) .

Your task is to process q queries of the form: how many trees are inside a given rectangle in the forest?

Constraints

- $1 \leq n \leq 1000$
- $1 \leq q \leq 2 \cdot 10^5$
- $1 \leq y1 \leq y2 \leq n$
- $1 \leq x1 \leq x2 \leq n$

Input

The first input line has two integers n and q : the size of the forest and the number of queries.

Then, there are n lines describing the forest. Each line has n characters: $.$ is an empty square and $*$ is a tree.

Finally, there are q lines describing the queries. Each line has four integers $y1, x1, y2, x2$ corresponding to the corners of a rectangle.

Output

Print the number of trees inside each rectangle.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4 3
.*..
*.**
**..
****
2 2 3 4
3 1 3 1
1 1 2 2
```

EXPECTED OUTPUT

```
3
1
2
```

Test Case 2

INPUT (STDIN)

```
4 3
.*.*
*.**
**.*
***.
1 2 3 4
3 2 3 4
1 4 2 3
```

EXPECTED OUTPUT

```
6
2
0
```

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int n, q;
6
7      scanf("%d %d", &n, &q);
8
9      char forest[n][n + 1];
10
11     // Read the forest
12     for (int i = 0; i < n; i++) {
13         scanf("%s", forest[i]);
14     }
15
16     // Prefix sum array
17     int prefix[n + 1][n + 1];
18
19     // Initialize
20     for (int i = 0; i <= n; i++) {
21         for (int j = 0; j <= n; j++) {
22             prefix[i][j] = 0;
23         }
24     }
25
26     // Build prefix sum
27     for (int i = 1; i <= n; i++) {
28         for (int j = 1; j <= n; j++) {
```

```

29         int tree = (forest[i - 1][j - 1] == '*');
30
31         prefix[i][j] =
32             tree
33             + prefix[i - 1][j]
34             + prefix[i][j - 1]
35             - prefix[i - 1][j - 1];
36     }
37 }
38
39 // Process queries
40 for (int i = 0; i < q; i++) {
41
42     int x1, y1, x2, y2;
43
44     scanf("%d %d %d %d", &x1, &y1, &x2, &y2);
45
46     int answer =
47         prefix[x2][y2]
48         - prefix[x1 - 1][y2]
49         - prefix[x2][y1 - 1]
50         + prefix[x1 - 1][y1 - 1];
51
52     printf("%d\n", answer);
53 }
54 return 0;
55 }

```

Output:

Test Case-1:

```

4 3
.*..
*.*
**..
****
2 2 3 4
3
3 1 3 1
1
1 1 2 2
2

```

Test Case-2:

	4 3	
	. * . *	
	* . **	
	** . *	
	*** .	
	1 2 3 4	
	6	
	3 2 3 4	
	2	
	1 4 2 3	
	0	

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